



Presenting Data

Presenting data

- Data is typically presented in the following forms:
- Tables
- Line graphs
- Bar charts
- Pie charts
- Histograms



Tables

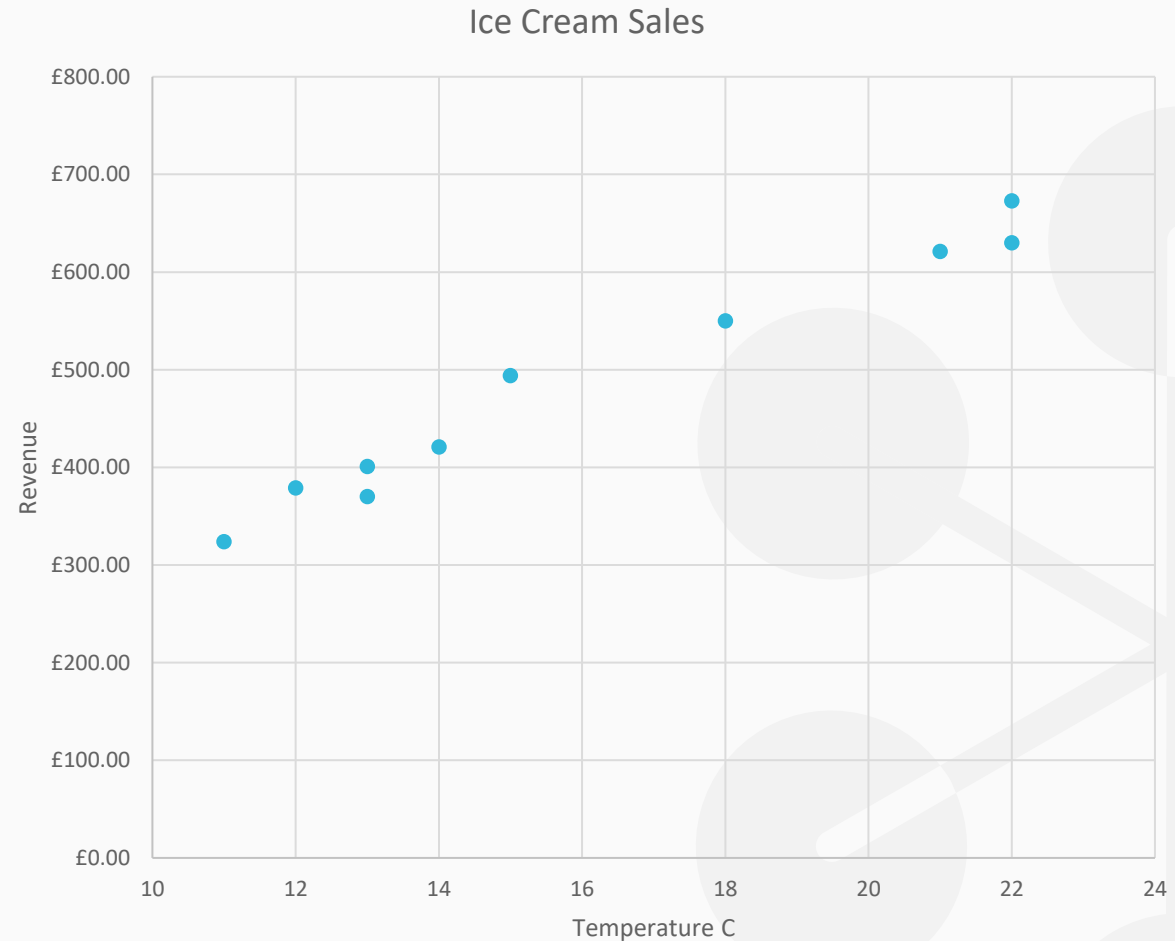
- Tables show the data in text and number form.
- This is useful when you are concerned with the exact figures.
- It is less effective when you want to see relationships, trends, patterns, proportion and so on.
- Graphs or charts are more effective at presenting data visually.
- Note that the terms “graph” and “chart” tend to be used interchangeably.

This table shows the revenue generated by ice cream sales against temperature over 10 days

Ice Cream Sales			
Day	Temperature		Revenue
1	18		£550.00
2	13		£370.00
3	21		£62.00
4	22		£67.00
5	22		£63.00
6	14		£42.00
7	12		£37.00
8	11		£32.00
9	15		£49.00
10	13		£40.00

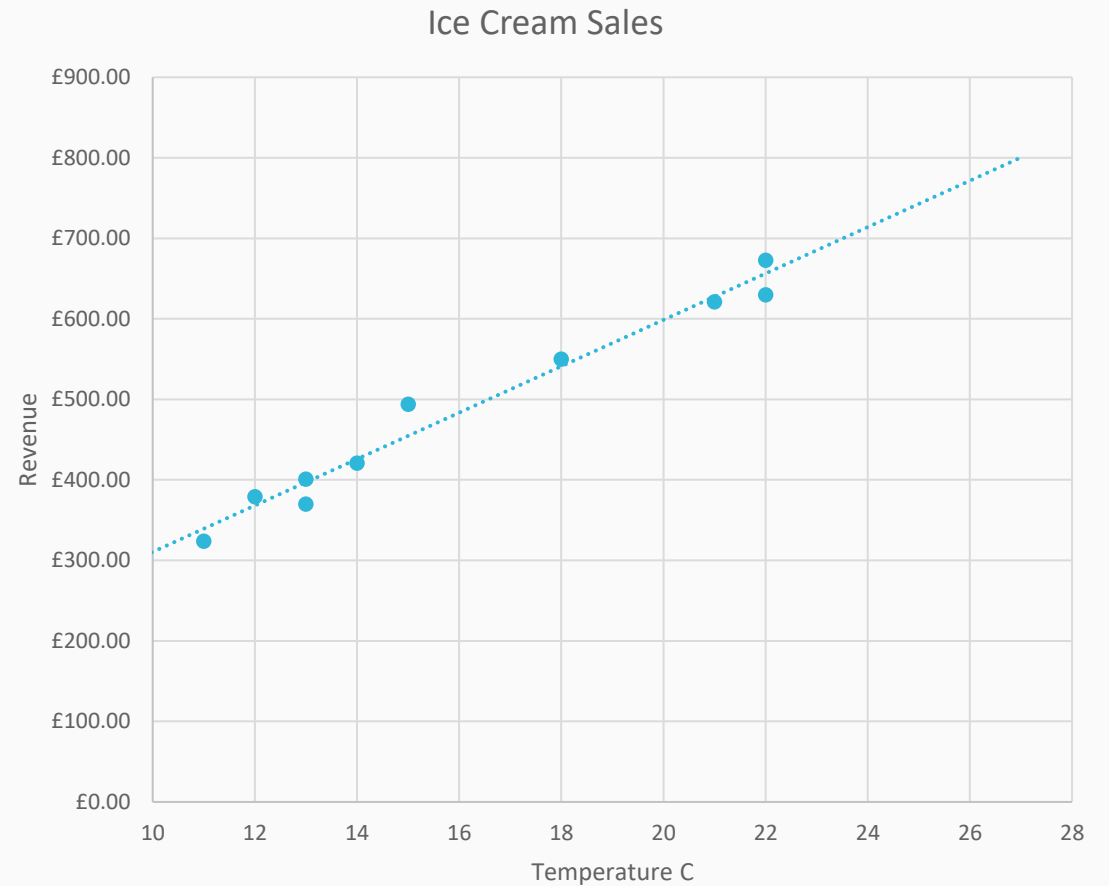
This data can be presented as a scatter chart

Ice Cream Sales		
Day	Temperature	Revenue
1	18	£550.00
2	13	£370.00
3	21	£621.00
4	22	£673.00
5	22	£630.00
6	14	£421.00
7	12	£379.00
8	11	£324.00
9	15	£494.00
10	13	£401.00



Purpose of scatter chart

- A scatter chart's purpose is to see if there is a relationship between two quantities.
- In this example, there is a distinct positive correlation between revenue and temperature.
- The trend line highlights this.

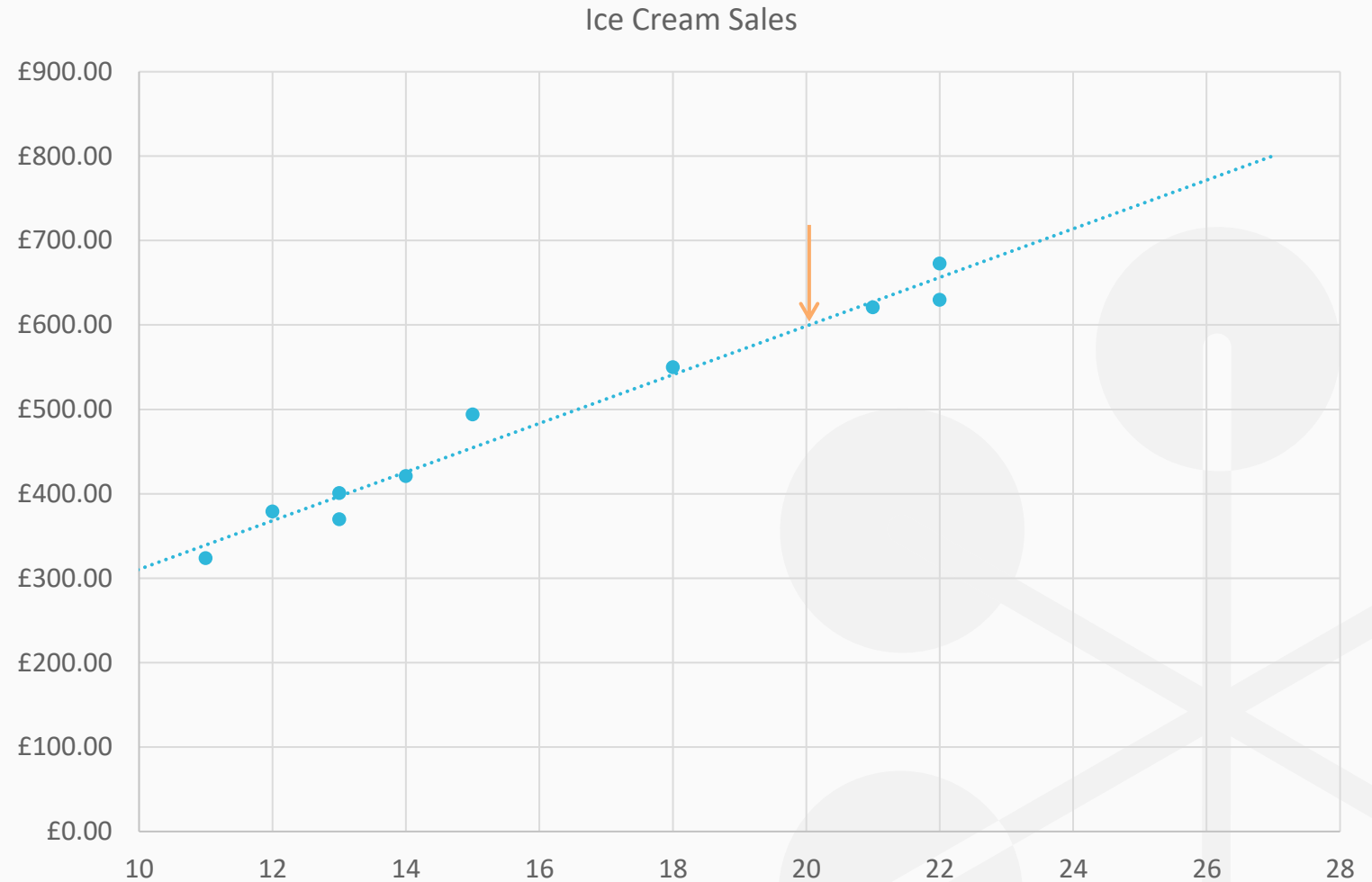


Extrapolation and interpolation

- The trend line allows us to make predictions.
- However, this must be done with caution!
- We cannot know for certain how far the trend applies beyond the set of data.
- And even if the trend is well established, there may be other reasons why the prediction does not come true.

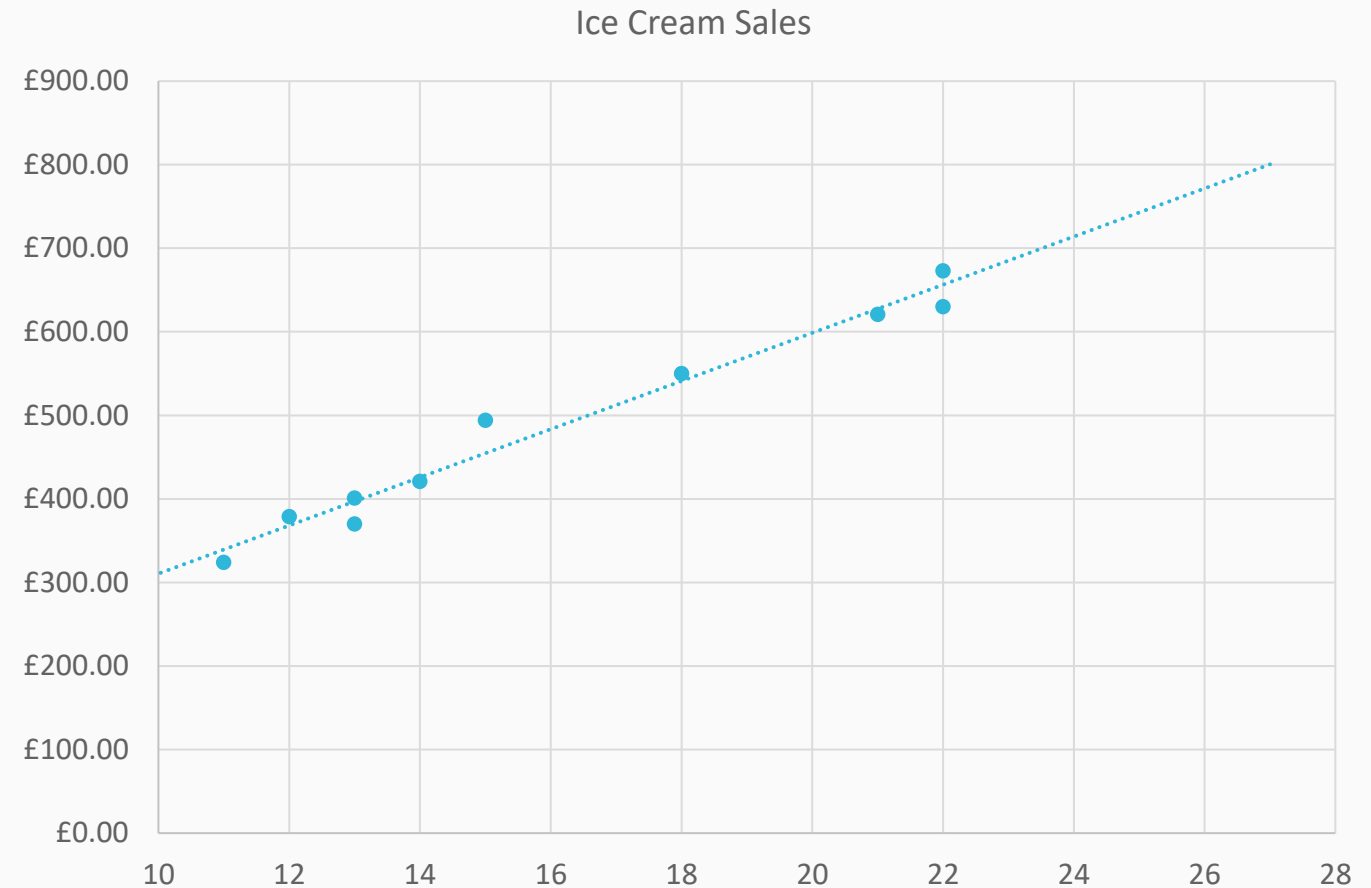
Interpolation

- This is between data points.
- It is reasonable to expect sales of around £600 when the temperature is 20C.
- (Realistically, if we are to be confident in our prediction, we would need more data points than given in this example.)



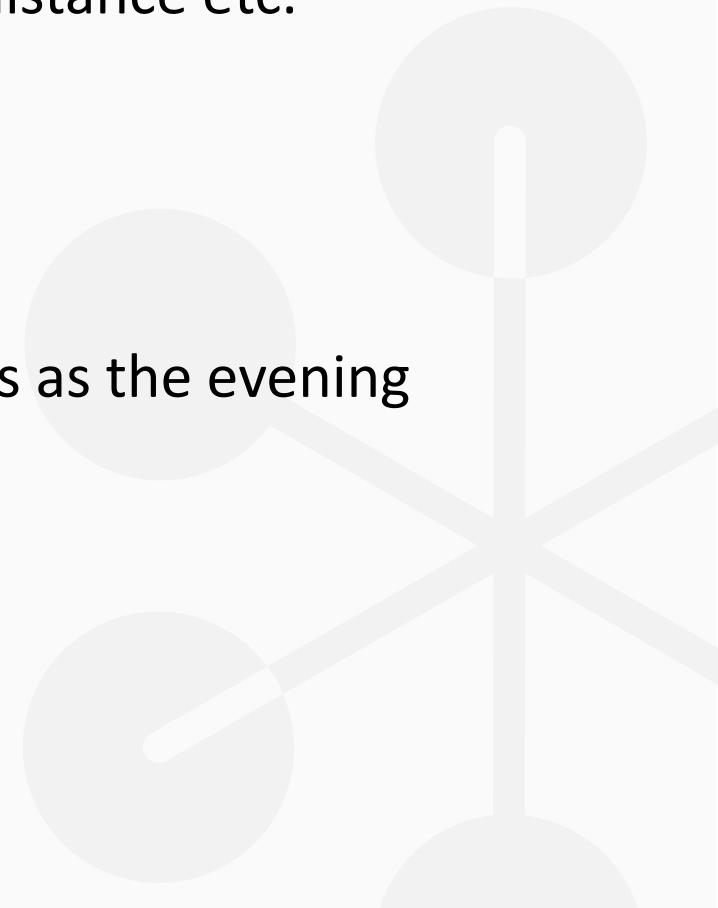
Extrapolation

- This is outside the data points.
- We cannot predict (with confidence) that a temperature of 27C will result in revenue of £800.
- This is because we cannot be sure how far the trend continues.
- The same is true for predictions below 10C.



Continuous data

- Line graphs are used to track progressions or changes over time, distance etc.
- Successive data points are linked together.
- In the example on the next slide, we can see that temperature falls as the evening progresses, as we would expect.



Presenting data

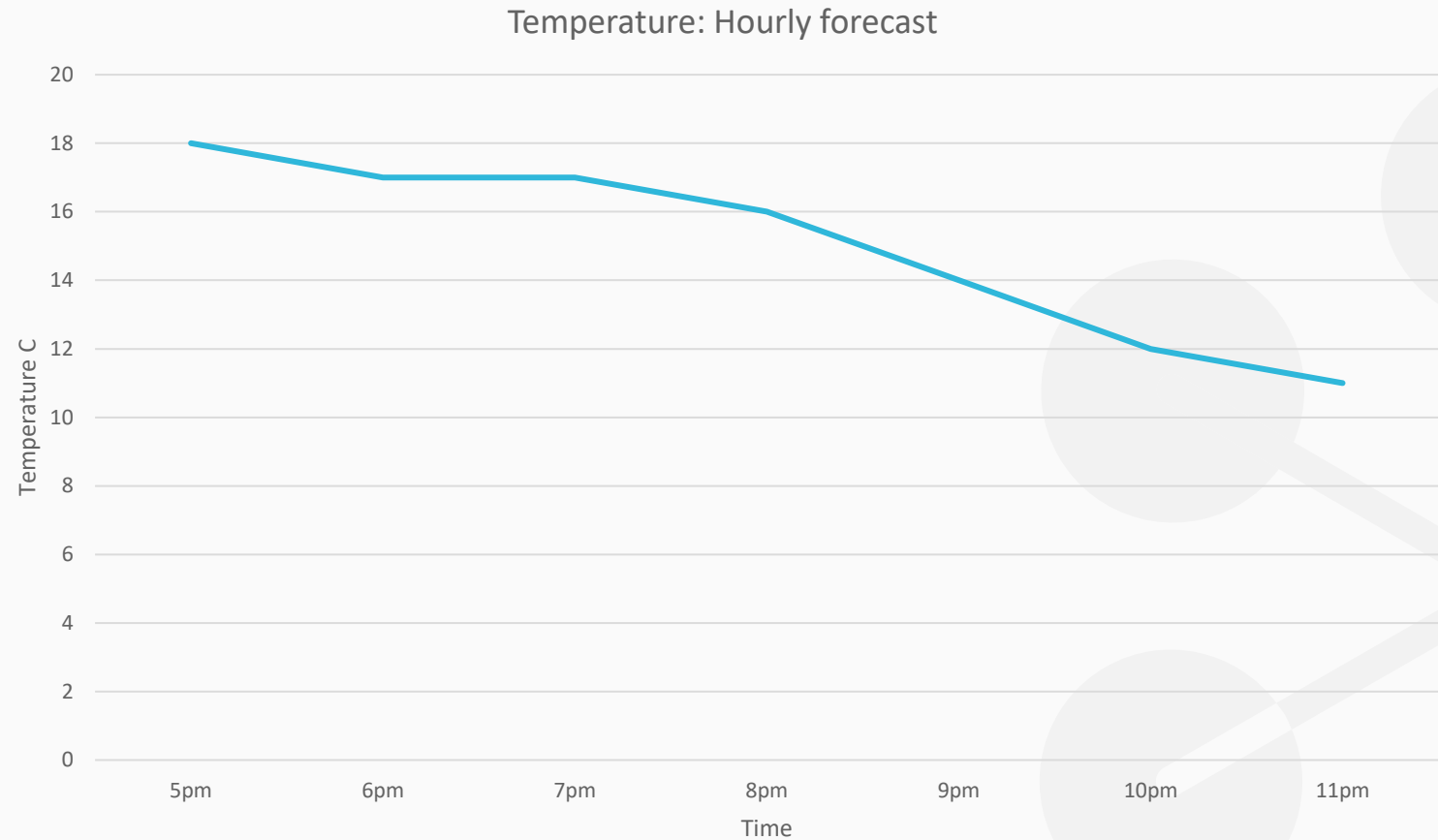
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Line graph

Temperature: Hourly forecast

Time	Temperature (C)
5pm	18
6pm	17
7pm	17
8pm	16
9pm	14
10pm	12
11pm	11



Discrete data

- When dealing with discrete (i.e. not continuous) categories, it does not make sense to use a line graph.

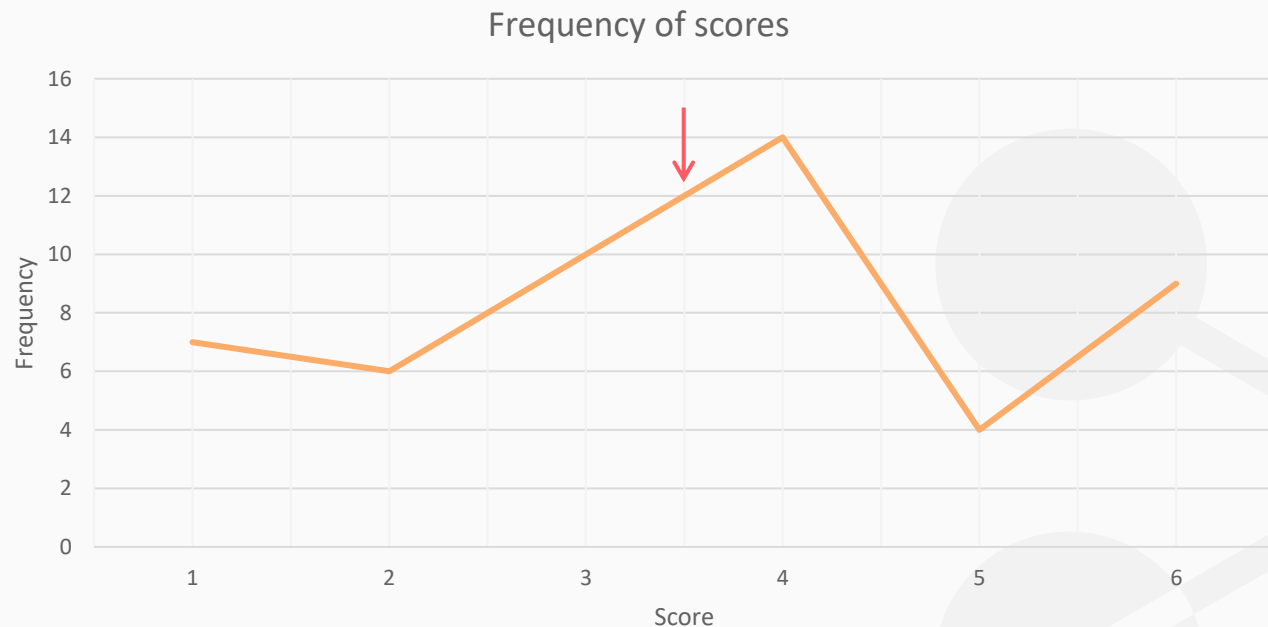
City populations	
City	Population
Aberdeen	231780
Edinburgh	530680
Glasgow	650300

Dice throws	
Score	Frequency
1	7
2	3
3	10
4	14
5	4
6	5

Unsuitable line graphs

The graph suggests that a score of $3\frac{1}{2}$ came up 12 times, which is of course nonsensical!

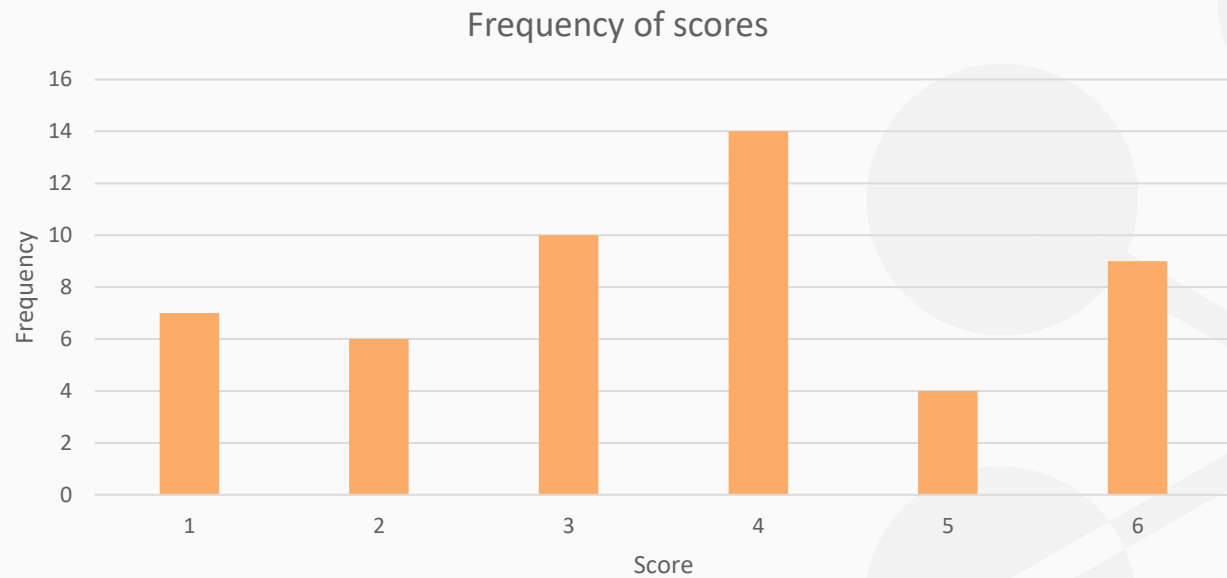
Dice throws	
Score	Frequency
1	7
2	6
3	10
4	14
5	4
6	9



Bar graphs

- A bar graph is a suitable means of visualising the dice data.

Dice throws	
Score	Frequency
1	7
2	6
3	10
4	14
5	4
6	9



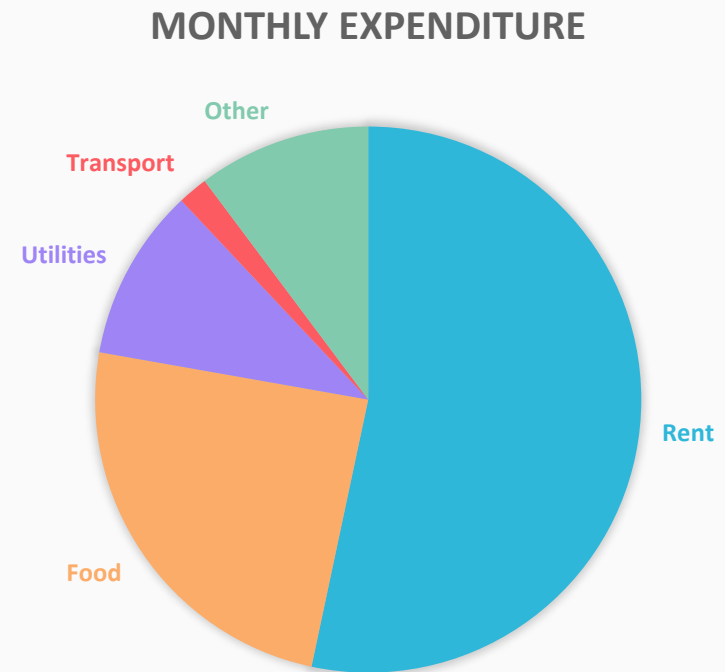
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Pie Charts

- Pie charts are used to highlight shares or proportions.

Expenses, budget of £2250 per month	
Item	Expenditure
Rent	£1,200.00
Food	£550.00
Utilities	£230.00
Transport	£40.00
Other	£230.00

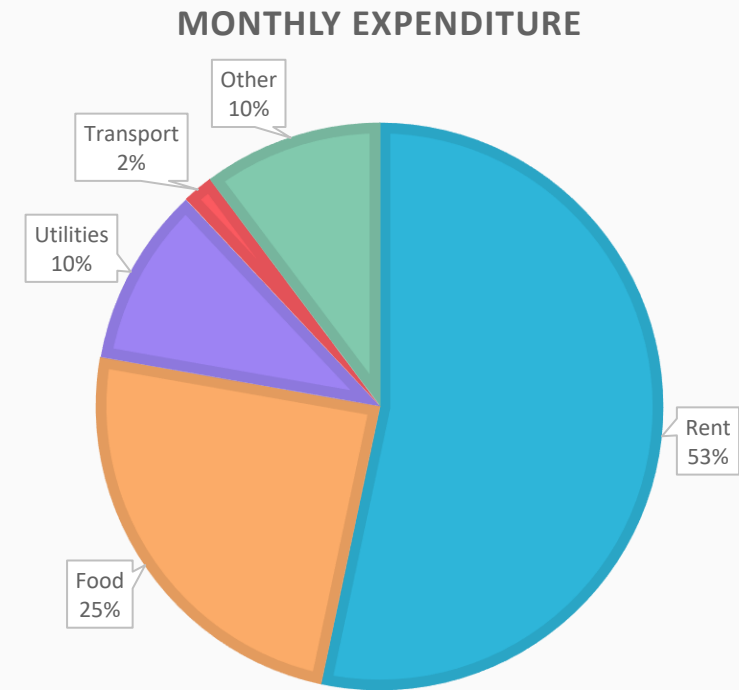
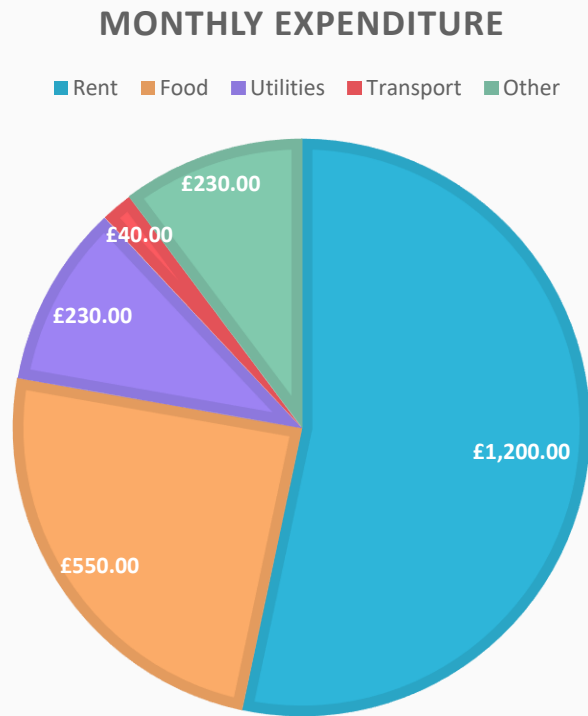


Pie Charts

- Pie charts show the proportions of an overall total, such as a budget. The slices should add up to 100% of that total.
- As with a real pie, if someone takes a larger slice, there is less for everyone else to share.
- Pie charts are best used for an overall impression. The human eye is not well suited to getting exact figures from looking at angles.
- Don't have too many slices – maybe have a maximum of 7. Otherwise, the chart will have some very thin slices which are hard to read. Some quantities may need to be grouped under “Other” or “Miscellaneous”.

Pie Charts

- The actual data may be displayed in different ways:



Histograms

- Histograms are similar to bar charts, but with a crucial difference.
- The data represented by each bar is grouped.
- In this example there are ages in the 20s, 30s and 40s.
- Each is added to a bin of width 10.

